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Percentile Rankings and Position Differences for Absolute and Allometrically Scaled Performance Measures From the National Football League Scouting Combine

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Abstract

Gillen, ZM. Percentile rankings and position differences for absolute and allometrically scaled performance measures from the National Football League scouting combine. *J Strength Cond Res* XX(X): 000–000, 2023—This study aimed to provide percentile rankings for absolute and allometrically scaled performance measures for National Football League (NFL) Scouting Combine subjects and determine if allometric scaling affected position-specific differences in performance measures. Subjects included American football players (n = 3,015) who participated in the NFL Scouting Combines between 2015 and 2019. Subjects were divided into position groups: defensive backs (DBs, n = 562), defensive linemen (DL, n = 498), linebackers (LBs, n = 395), offensive linemen (OL, n = 505), running backs (RBs, n = 303), tight ends (TEs, n = 165), quarterbacks (QBs, n = 165), and wide receivers (WRs, n = 422). Performance measures included 40-yd dash time (with 10- and 20-yd split times), bench press repetitions to failure, vertical jump height, broad jump distance, pro-agility time, and L-cone drill time. Descriptive statistics were computed for all performance metrics and allometrically scaled performance metrics. One-way analyses of variance tested for position group differences for absolute and allometrically scaled performance measures. Percentile rankings for absolute and allometrically scaled performance measures were determined. Position-specific differences for absolute performance measures indicated that DB and WR tended to be faster, jump higher and further, and have greater change-of-direction capabilities compared with LB, QB, RB, and TE, who scored better in these respects than OL and DL. Allometric scaling revealed that DL, LB, RB, and TE tended to have superior performance than DB and WR, with QB and OL having the poorest performance. This study provides percentile rankings and the equation and parameters by which coaches and practitioners may allometrically scale data for evaluations of normalized performance measures from the NFL Scouting Combine.

Key Words: American football, normative values, athletic performance, normalization

Introduction

The most popular athletic performance combine in the United States is the National Football League (NFL) Scouting Combine. This combine allows NFL scouts and coaches to assess athletic performance of potential players for success in the NFL and includes a battery of tests: 40-yd dash (with 10- and 20-yd splits), bench press repetitions to failure test, vertical jump (VJ), broad jump (BJ), pro agility (PA), and L-cone (LC) drills to assess speed, strength, power, and agility. It is believed that better combine scores may be indicative of greater playing potential, although this is not always the case (3,10,27,32). Nevertheless, it stands to reason that a battery of athletic performance tests at the very least allows NFL coaches and scouts a measuring stick by which to assess potential players.

One valuable method by which NFL Scouting Combine subjects may be assessed is by comparing the scores they achieve during the combine to normative percentile rankings from previous NFL Scouting Combines. Previous studies and texts have provided these metrics for various populations (8,8,14,15); however, because NFL Scouting Combine subjects can be considered elite-level American football players, it is important to compare their combine results with athletes of the same caliber. Nuzzo (20) is the only previous study to provide percentile ranking reference values for NFL Scouting Combine subjects using data from the 1999–2014 NFL Scouting Combines to provide coaches and practitioners with valuable rankings by which combine subjects may be compared. However, Fitzgerald and Jensen (5) recently reported that NFL Scouting Combine measures statistically changed when comparing data from 1999–2000 to 2015–2016, such that players in more recent combines performed better than players in previous years. Thus, it may be valuable to provide updated percentile rankings over a narrower window of time that represents the current generation of NFL Scouting Combine subjects. Furthermore, Nuzzo (20), in conjunction with other studies of NFL Scouting Combine data (27,29), has suggested that simply assessing absolute performance metrics obtained during these combines may not accurately paint the entire picture of athletic performance because body mass can often be a confounding factor. Therefore, previous studies have suggested using normalization procedures, namely, allometric scaling, to account for the influence of body mass on athletic performance among NFL Scouting Combine subjects (20,27,29).

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Table 1

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) 10-yd split (in seconds).*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	1.87	1.66	1.87	1.74	1.96	1.72	1.77	1.80	1.67
10	1.83	1.64	1.84	1.71	1.93	1.69	1.76	1.76	1.65
15	1.80	1.63	1.81	1.70	1.91	1.67	1.73	1.75	1.64
20	1.77	1.62	1.80	1.69	1.89	1.66	1.72	1.73	1.63
25	1.74	1.61	1.79	1.68	1.87	1.64	1.71	1.71	1.62
30	1.71	1.61	1.77	1.67	1.86	1.64	1.70	1.71	1.61
35	1.69	1.60	1.76	1.65	1.85	1.63	1.69	1.70	1.60
40	1.67	1.59	1.75	1.65	1.84	1.62	1.68	1.69	1.60
45	1.65	1.59	1.74	1.64	1.84	1.62	1.68	1.69	1.59
50	1.64	1.58	1.73	1.64	1.83	1.61	1.67	1.68	1.58
55	1.63	1.58	1.71	1.63	1.82	1.60	1.66	1.67	1.58
60	1.62	1.57	1.70	1.62	1.81	1.59	1.65	1.67	1.57
65	1.61	1.57	1.69	1.61	1.80	1.58	1.65	1.65	1.57
70	1.60	1.56	1.68	1.61	1.79	1.58	1.63	1.65	1.56
75	1.59	1.55	1.67	1.60	1.78	1.57	1.63	1.64	1.55
80	1.58	1.54	1.65	1.59	1.77	1.56	1.62	1.62	1.54
85	1.56	1.53	1.64	1.58	1.76	1.55	1.61	1.61	1.53
90	1.55	1.52	1.63	1.57	1.73	1.54	1.59	1.59	1.52
95	1.53	1.51	1.60	1.55	1.71	1.52	1.57	1.57	1.51
Mean	1.67	1.58^{2-7}	$1.73^{1,3-8}$	$1.64^{1,2,4-8}$	$1.83^{1-3,5-8}$	$1.61^{1-4,6-8}$	$1.67^{1-5,8}$	$1.68^{1-5,8}$	1.59^{2-7}
SD	0.11	0.05	0.08	0.06	0.07	0.06	0.06	0.07	0.05
n	2,752	525	454	355	443	284	151	146	394
Percentile									
5	0.476	0.472	0.469	0.470	0.487	0.470	0.470	0.490	0.472
10	0.469	0.466	0.463	0.461	0.479	0.464	0.465	0.484	0.465
15	0.464	0.463	0.458	0.457	0.475	0.459	0.458	0.477	0.463
20	0.461	0.460	0.455	0.452	0.471	0.456	0.455	0.475	0.459
25	0.458	0.457	0.452	0.449	0.466	0.453	0.451	0.473	0.457
30	0.456	0.455	0.450	0.447	0.464	0.450	0.449	0.471	0.456
35	0.453	0.454	0.448	0.444	0.461	0.448	0.446	0.469	0.453
40	0.451	0.452	0.445	0.442	0.460	0.445	0.444	0.466	0.451
45	0.449	0.450	0.443	0.441	0.458	0.444	0.443	0.462	0.449
50	0.446	0.448	0.441	0.439	0.456	0.442	0.442	0.461	0.447
55	0.444	0.446	0.439	0.438	0.455	0.441	0.440	0.458	0.445
60	0.442	0.444	0.437	0.435	0.453	0.440	0.438	0.456	0.442
65	0.440	0.442	0.435	0.434	0.451	0.438	0.437	0.455	0.441
70	0.438	0.440	0.433	0.432	0.448	0.436	0.433	0.452	0.439
75	0.435	0.439	0.429	0.430	0.446	0.433	0.431	0.449	0.436
80	0.432	0.436	0.427	0.428	0.442	0.431	0.430	0.444	0.434
85	0.430	0.433	0.424	0.424	0.439	0.429	0.427	0.441	0.431
90	0.426	0.430	0.422	0.421	0.433	0.426	0.420	0.437	0.428
95	0.421	0.425	0.416	0.416	0.429	0.421	0.413	0.430	0.422
Mean	0.447	0.448^{2-7}	$0.442^{1,4,7,8}$	$0.440^{1,4,7,8}$	$0.457^{1-3,5-8}$	$0.443^{1,4,7}$	$0.442^{1,4,7,8}$	$0.461^{1-6,8}$	$0.447^{2-4,6,7}$
SD	0.017	0.014	0.016	0.016	0.017	0.015	0.016	0.018	0.015
n	2,752	525	454	355	443	284	151	146	394

*[1]DB = defensive back; [2]DL = defensive linemen; [3]LB = linebacker; [4]OL = offensive linemen; [5]RB = running back; [6]TE = tight end; [7]QB = quarterback; [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

The purpose of allometric scaling is to scale a variable to an individual's body mass to an exponential power called the allometric parameter. This allometric parameter differs depending on the chosen test, allowing the flexibility for researchers and practitioners to scale to a specific performance test. Previous studies have advocated for the use of allometric scaling to normalize athletic performance metrics over the commonly used ratio scaling method (9,12,17,18,20), which simply involves dividing absolute performance by body mass. However, an underlying assumption of ratio scaling is that the model of best fit between 2 variables is a linear model (9,17,18). If this is not the case, then one could argue that ratio scaling is inappropriate, which makes the argument that allometric scaling may be a superior normalization procedure. Previous studies using data from the NFL Scouting Combine have agreed with this notion (20,27,29), suggesting that future studies use allometric scaling as a normalization procedure to allow more accurate comparisons of players with different body masses and in different position groups. In fact, Nuzzo (20) suggested that future studies should not only use allometric scaling in lieu of ratio scaling but also should consider calculating percentile rankings for allometrically scaled performance measures, which may provide coaches and practitioners a more holistic picture of true athletic performance. Furthermore, although the number of studies examining position-specific differences among NFL Scouting Combine performance measures is limited (5,7), position-specific comparisons using allometrically scaled data may provide unique insight regarding the underlying factors affecting differences in performance across playing positions in American football. Therefore, the purpose of this study was twofold. First, this study aimed to provide updated percentile

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Table 2

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) 20-yd split time (in seconds).*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	3.11	2.74	3.09	2.87	3.22	2.82	2.91	2.95	2.74
10	3.04	2.71	3.04	2.83	3.18	2.78	2.88	2.91	2.72
15	2.99	2.70	3.01	2.82	3.15	2.75	2.86	2.89	2.70
20	2.93	2.69	2.99	2.80	3.13	2.74	2.85	2.88	2.69
25	2.88	2.68	2.96	2.78	3.11	2.72	2.84	2.86	2.68
30	2.84	2.67	2.94	2.77	3.10	2.71	2.82	2.84	2.66
35	2.80	2.66	2.91	2.76	3.08	2.70	2.81	2.84	2.65
40	2.77	2.65	2.90	2.75	3.07	2.69	2.79	2.81	2.65
45	2.75	2.64	2.88	2.73	3.06	2.68	2.78	2.80	2.64
50	2.72	2.64	2.87	2.72	3.04	2.67	2.77	2.80	2.63
55	2.71	2.63	2.85	2.71	3.03	2.66	2.76	2.79	2.63
60	2.69	2.62	2.84	2.70	3.02	2.65	2.75	2.77	2.62
65	2.68	2.61	2.82	2.69	3.01	2.64	2.74	2.76	2.61
70	2.66	2.60	2.81	2.68	3.00	2.63	2.73	2.75	2.60
75	2.64	2.59	2.78	2.67	2.98	2.62	2.72	2.74	2.59
80	2.63	2.58	2.77	2.66	2.96	2.61	2.70	2.72	2.58
85	2.61	2.56	2.75	2.65	2.94	2.59	2.70	2.70	2.57
90	2.59	2.55	2.72	2.63	2.90	2.57	2.66	2.69	2.55
95	2.56	2.53	2.68	2.60	2.86	2.54	2.64	2.65	2.52
Mean	2.77	2.63^{2-7}	$2.87^{1,3-8}$	$2.73^{1,2,4-8}$	$3.05^{1-3,5-8}$	$2.67^{1-4,6-8}$	$2.77^{1-5,8}$	$2.80^{1-5,8}$	2.63^{2-7}
SD	0.17	0.06	0.13	0.08	0.11	0.08	0.08	0.09	0.08
n	2,752	525	454	355	443	284	151	146	394
Percentile									
5	0.781	0.776	0.768	0.765	0.796	0.768	0.766	0.804	0.774
10	0.771	0.766	0.758	0.757	0.786	0.758	0.758	0.792	0.766
15	0.764	0.762	0.751	0.752	0.779	0.752	0.754	0.788	0.761
20	0.759	0.759	0.748	0.747	0.774	0.748	0.748	0.783	0.756
25	0.755	0.754	0.744	0.743	0.769	0.746	0.741	0.781	0.753
30	0.751	0.751	0.741	0.741	0.767	0.743	0.740	0.779	0.749
35	0.748	0.749	0.739	0.737	0.764	0.739	0.737	0.775	0.747
40	0.745	0.747	0.736	0.733	0.761	0.737	0.734	0.771	0.744
45	0.742	0.744	0.733	0.729	0.757	0.734	0.732	0.768	0.740
50	0.738	0.742	0.729	0.727	0.756	0.731	0.730	0.765	0.737
55	0.735	0.738	0.727	0.725	0.754	0.729	0.727	0.762	0.735
60	0.731	0.736	0.725	0.722	0.750	0.727	0.725	0.757	0.731
65	0.728	0.733	0.722	0.720	0.748	0.724	0.721	0.755	0.729
70	0.726	0.730	0.719	0.716	0.744	0.722	0.718	0.749	0.726
75	0.722	0.728	0.715	0.712	0.740	0.720	0.716	0.744	0.722
80	0.719	0.725	0.711	0.710	0.735	0.716	0.713	0.741	0.719
85	0.714	0.722	0.706	0.707	0.728	0.714	0.710	0.737	0.716
90	0.710	0.718	0.702	0.703	0.720	0.709	0.702	0.729	0.710
95	0.701	0.712	0.695	0.697	0.713	0.701	0.692	0.719	0.701
Mean	0.025	0.742^{2-7}	$0.730^{1,4,7,8}$	$0.729^{1,4,7,8}$	$0.755^{1-3,5-8}$	$0.733^{1,4,7}$	$0.730^{1,4,7,8}$	$0.763^{1-6,8}$	$0.737^{2-4,6,7}$
SD	0.0005	0.019	0.025	0.023	0.025	0.020	0.021	0.025	0.026
n	2,752	525	454	355	443	284	151	146	394

*[1]DB = defensive back; [2]DL = defensive linemen; [3]LB = linebacker; [4]OL = offensive linemen; [5]RB = running back; [6]TE = tight end; [7]QB = quarterback; [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

rankings for absolute performance measures for NFL Scouting Combine subjects as a whole and split into position groups. Second, this study aimed to provide percentile rankings for allometrically scaled performance measures for NFL Scouting Combine subjects as a whole and split into position groups and determine if allometric scaling affected position-specific differences in athletic performance.

Methods

Experimental Approach to the Problem

This study analyzed NFL Scouting Combine performance data between 2015 and 2019. Because of the COVID-19 pandemic, data from 2020 to 2021 were excluded. Furthermore, because of the drastic reduction in subjects and modifications as a result of the COVID-19 pandemic, data from 2022 were excluded from the present investigation. Therefore, data from 2015 to 2019 were chosen to provide a representation of the current generation of NFL Scouting Combine subjects. These data were used to determine overall absolute and allometrically scaled percentile rank scores for all subjects, as well as percentile rank scores based on position groups and to compare differences in absolute and allometrically scaled performance among position groups. Therefore, the independent variable in this study was position group, whereas the dependent variables were 40-yd dash time (with 10- and 20-yd split times), bench press repetitions to failure, VJ height, BJ distance, PA time, and LC drill time.

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Table 3

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) 40-yd dash time (in seconds).*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	5.38	4.69	5.37	4.95	5.57	4.88	5.00	5.12	4.69
10	5.27	4.66	5.28	4.89	5.51	4.80	4.94	5.06	4.65
15	5.18	4.63	5.21	4.85	5.46	4.76	4.91	5.03	4.63
20	5.08	4.62	5.16	4.83	5.43	4.73	4.89	4.95	4.62
25	4.98	4.60	5.13	4.80	5.39	4.68	4.87	4.93	4.59
30	4.91	4.59	5.10	4.78	5.37	4.66	4.84	4.91	4.58
35	4.84	4.57	5.06	4.77	5.34	4.64	4.82	4.90	4.56
40	4.80	4.56	5.02	4.75	5.32	4.62	4.80	4.87	4.55
45	4.75	4.55	5.00	4.73	5.30	4.60	4.79	4.84	4.54
50	4.70	4.54	4.97	4.72	5.27	4.59	4.78	4.84	4.54
55	4.67	4.53	4.94	4.70	5.26	4.57	4.76	4.81	4.53
60	4.63	4.52	4.92	4.68	5.23	4.56	4.74	4.80	4.51
65	4.60	4.50	4.89	4.66	5.22	4.54	4.73	4.79	4.50
70	4.58	4.49	4.87	4.65	5.19	4.52	4.72	4.78	4.48
75	4.55	4.48	4.84	4.62	5.17	4.50	4.70	4.76	4.46
80	4.53	4.46	4.82	4.60	5.14	4.49	4.68	4.74	4.44
85	4.50	4.44	4.76	4.58	5.09	4.47	4.65	4.69	4.42
90	4.47	4.42	4.72	4.55	5.03	4.45	4.62	4.64	4.39
95	4.42	4.37	4.64	4.50	4.96	4.40	4.56	4.60	4.36
Mean	4.79	4.54^{2-7}	$4.99^{1,3-8}$	$4.72^{1,2,4-8}$	$5.28^{1-3,5-8}$	$4.61^{1-4,6-8}$	$4.78^{1-5,7,8}$	$4.85^{1-6,8}$	4.53^{2-7}
SD	0.31	0.10	0.22	0.14	0.18	0.14	0.13	0.15	0.10
n	2,845	544	468	365	460	291	156	154	407
Percentile									
5	1.273	1.255	1.250	1.244	1.296	1.242	1.240	1.309	1.260
10	1.253	1.246	1.236	1.230	1.283	1.232	1.224	1.299	1.243
15	1.243	1.237	1.226	1.218	1.268	1.223	1.215	1.292	1.234
20	1.234	1.233	1.219	1.213	1.260	1.213	1.211	1.282	1.227
25	1.227	1.227	1.211	1.208	1.255	1.209	1.206	1.276	1.220
30	1.221	1.221	1.205	1.203	1.248	1.204	1.199	1.268	1.215
35	1.215	1.218	1.199	1.200	1.243	1.200	1.195	1.260	1.209
40	1.210	1.214	1.196	1.195	1.238	1.197	1.190	1.255	1.204
45	1.205	1.210	1.190	1.190	1.233	1.193	1.185	1.253	1.200
50	1.200	1.206	1.187	1.188	1.228	1.189	1.181	1.248	1.195
55	1.196	1.203	1.183	1.181	1.223	1.183	1.178	1.245	1.191
60	1.191	1.200	1.179	1.178	1.219	1.181	1.175	1.237	1.187
65	1.186	1.196	1.174	1.173	1.214	1.177	1.170	1.234	1.184
70	1.181	1.192	1.170	1.169	1.209	1.172	1.166	1.227	1.179
75	1.175	1.186	1.165	1.163	1.205	1.169	1.160	1.221	1.174
80	1.169	1.181	1.160	1.159	1.197	1.166	1.158	1.209	1.169
85	1.164	1.174	1.153	1.154	1.190	1.161	1.154	1.200	1.164
90	1.156	1.168	1.146	1.145	1.178	1.151	1.149	1.192	1.155
95	1.144	1.158	1.131	1.133	1.159	1.139	1.132	1.181	1.146
Mean	1.203	1.207^{2-8}	$1.189^{1,4,7,8}$	$1.187^{1,4,7,8}$	$1.229^{1-3,5-8}$	$1.190^{1,4,7}$	$1.184^{1,4,7,8}$	$1.247^{1-6,8}$	$1.198^{1-4,6,7}$
SD	0.039	0.030	0.036	0.034	0.039	0.030	0.031	0.041	0.035
n	2,845	544	468	365	460	291	156	154	407

*[1]DB = defensive back; [2]DL = defensive linemen; [3]LB = linebacker; [4]OL = offensive linemen; [5]RB = running back; [6]TE = tight end; [7]QB = quarterback; [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

Subjects

Subjects included American football players (n = 3,015) who participated in the NFL Scouting Combines between 2015 and 2019. Subjects were further divided into the following position groups: defensive backs (DBs, n = 562), defensive linemen (DL, n = 498), linebackers (LBs, n = 395), offensive linemen (OL, n = 505), running backs (RBs, n = 303), tight ends (TEs, n = 165), quarterbacks (QBs, n = 165), and wide receivers (WRs, n = 422). The Mississippi State University Institutional Review Board (IRB) determined the data were considered exempt (official letter from the IRB, November 9, 2021) because the data are accessible publicly (19).

Procedures

Anthropometrics. For measurements of height and body mass, subjects were required to wear compression garments and no shoes or cleats. Height was recorded in centimeters, and body mass was recorded in kilograms.

40-Yd Dash. For the 40-yd dash, subjects began in a 3-point sprinters stance with their hand on the ground. Subjects then sprinted straight forward as fast as possible through the finish line marked 40 yds from the starting line. The test began when the subject's hand broke contact with the ground and ended when the subject passed through the finish line. Split times were also

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Table 4

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) bench press repetitions to failure.*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	10.00	8.00	17.00	15.00	16.00	11.50	12.00	—	8.00
10	12.00	9.00	18.00	15.00	17.00	14.00	14.00	—	9.00
15	13.00	10.00	19.95	16.00	19.00	15.00	15.00	—	10.00
20	14.00	11.00	20.00	17.00	20.00	15.00	15.00	—	11.00
25	15.00	12.00	21.00	18.00	21.00	16.00	16.00	—	11.00
30	16.00	13.00	22.00	18.30	21.00	17.00	17.00	—	12.00
35	17.00	13.00	22.00	19.00	22.00	17.00	17.00	—	12.00
40	18.00	14.00	23.00	19.00	23.00	18.00	18.00	—	13.00
45	19.00	14.00	23.00	20.00	23.00	19.00	18.00	—	13.00
50	19.00	15.00	24.00	20.00	24.00	19.00	19.00	—	14.00
55	20.00	15.00	24.15	21.00	24.00	19.00	19.00	—	14.00
60	21.00	16.00	25.00	22.00	25.00	20.00	20.00	—	15.00
65	22.00	16.00	26.00	23.00	26.00	21.00	20.00	—	15.00
70	23.00	17.00	26.00	23.00	26.00	21.00	21.00	—	16.00
75	24.00	18.00	28.00	24.00	27.00	22.00	22.00	—	17.00
80	25.00	18.00	28.00	24.00	28.00	23.00	22.00	—	17.00
85	26.00	19.00	30.00	25.00	30.00	24.00	23.00	—	18.00
90	28.00	20.00	31.00	26.00	32.00	25.00	24.00	—	19.00
95	30.00	21.00	34.00	29.00	34.00	27.50	25.00	—	21.00
Mean	19.64	14.79[2–6]	24.37[1,3,5,6,8]	20.85[1,2,4-6,8]	24.13[1,3,5,6,8]	19.29[1–4,8]	18.71[1–4,8]	—	13.97
SD	6.26	4.26	5.21	4.37	5.48	4.86	3.84	—	4.10
n	2,504	492	452	340	461	269	147	—	343
Percentile									
5	0.0372	0.0341	0.0432	0.0466	0.0363	0.0461	0.0377	—	0.0315
10	0.0424	0.0393	0.0475	0.0503	0.0404	0.0506	0.0422	—	0.0356
15	0.0458	0.0418	0.0497	0.0540	0.0436	0.0529	0.0445	—	0.0407
20	0.0487	0.0455	0.0526	0.0561	0.0458	0.0549	0.0466	—	0.0433
25	0.0514	0.0479	0.0549	0.0581	0.0478	0.0570	0.0487	—	0.0452
30	0.0533	0.0517	0.0571	0.0597	0.0501	0.0597	0.0515	—	0.0472
35	0.0552	0.0540	0.0586	0.0609	0.0517	0.0627	0.0524	—	0.0492
40	0.0572	0.0568	0.0598	0.0630	0.0531	0.0659	0.0537	—	0.0517
45	0.0587	0.0582	0.0612	0.0645	0.0546	0.0677	0.0546	—	0.0535
50	0.0603	0.0596	0.0626	0.0669	0.0556	0.0697	0.0564	—	0.0549
55	0.0623	0.0619	0.0638	0.0685	0.0572	0.0721	0.0582	—	0.0567
60	0.0642	0.0644	0.0660	0.0707	0.0586	0.0736	0.0604	—	0.0586
65	0.0662	0.0660	0.0676	0.0731	0.0601	0.0752	0.0618	—	0.0606
70	0.0682	0.0681	0.0693	0.0743	0.0620	0.0772	0.0631	—	0.0628
75	0.0709	0.0707	0.0715	0.0764	0.0638	0.0811	0.0651	—	0.0648
80	0.0736	0.0728	0.0732	0.0787	0.0666	0.0828	0.0673	—	0.0669
85	0.0764	0.0762	0.0758	0.0817	0.0694	0.0855	0.0689	—	0.0689
90	0.0808	0.0798	0.0797	0.0857	0.0744	0.0894	0.0716	—	0.0747
95	0.0868	0.0860	0.0835	0.0952	0.0807	0.0956	0.0744	—	0.0803
Mean	0.0612	0.0599[2–5,8]	0.0634[1,3-6,8]	0.0676[1,2,4,6,8]	0.0566[1–3,5,6]	0.0697[1,2,4,6,8]	0.0568[2,3,5,8]	—	0.0551[1–3,5,6]
SD	0.0153	0.0165	0.0129	0.0142	0.0131	0.0158	0.0115	—	0.0149
n	2,504	492	452	340	461	269	147	—	343

*[1]DB = defensive back; [2]DL = defensive linemen; [3]LB = linebacker; [4]OL = offensive linemen; [5]RB = running back; [6]TE = tight end; [7]QB = quarterback; [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

recorded at the 10- and 20-yd marks. These times were recorded using laser timing gates (Zybek Sports, Broomfield, CO).

Bench Press. For the bench press test, subjects were required to complete as many repetitions of the bench press exercise as possible with 225 lbs (102.3 kg). A repetition was counted if the subject lowered the bar all the way to the chest and pressed the bar until the elbows were fully extended. Subjects were required to maintain contact on the bench with their head, shoulders, and lower back or buttocks, and contact on the ground with their right and left foot. For QBs, only 8 subjects completed this test between 2015 and 2019; therefore, it was excluded for this position group for all subsequent analyses.

Vertical Jump. To perform the VJ, standing reach of each subject was measured to assess maximal reach and establish a baseline mark. Subjects then began in a ready position, performed a rapid countermovement of self-selected depth, followed by a maximal vertical jump. Jump height was considered the difference between maximal reach height and maximal jump height, recorded to the nearest 0.5 inch (1.27 cm). This assessment was measured using a jump device (e.g., Vertec).

Broad Jump. To perform the BJ, subjects began with their toes even with the starting line and then performed a rapid countermovement of self-selected depth, followed by a maximal horizontal jump as far as possible. Subjects were required to land with their feet planted firmly on the ground. Jump distance was

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Table 5

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) vertical jump height (in centimeters).*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	63.50	77.47	62.23	70.10	57.15	73.66	71.12	67.31	76.20
10	67.31	80.01	64.77	73.66	59.69	75.44	74.04	68.58	78.74
15	71.12	81.28	67.31	76.20	60.96	77.47	74.93	71.12	81.28
20	73.66	82.55	69.85	77.47	63.50	79.76	76.20	72.39	82.55
25	74.93	83.82	72.39	78.74	64.77	81.28	77.47	73.66	83.82
30	77.47	85.09	73.66	81.28	66.04	81.28	78.74	74.93	85.09
35	78.74	86.36	73.66	81.28	67.31	83.06	80.01	76.20	86.36
40	80.01	86.36	74.93	83.82	68.58	83.82	80.26	77.47	86.36
45	81.28	87.63	76.20	85.09	68.58	85.09	81.28	78.74	87.63
50	82.55	88.90	77.47	85.09	69.85	86.36	82.55	78.74	88.90
55	83.82	90.17	78.74	86.36	71.12	87.63	83.82	80.01	90.17
60	85.09	90.17	80.01	87.63	71.12	88.90	84.84	80.01	90.17
65	86.93	91.44	81.28	87.63	72.39	90.17	86.36	82.49	91.44
70	88.90	92.71	82.55	88.90	73.66	90.17	87.63	83.82	92.71
75	90.17	93.98	82.55	90.17	74.93	91.44	87.63	85.09	93.98
80	91.44	96.52	84.58	91.44	77.22	93.98	88.90	86.36	95.25
85	92.71	97.79	86.36	93.98	77.47	94.49	91.44	88.90	96.52
90	95.25	99.95	88.90	96.52	80.01	97.79	94.49	90.17	99.06
95	99.06	103.95	92.71	100.33	82.55	100.33	96.52	92.71	102.24
Mean	82.33	89.30[2–7]	77.55[1,3-6,8]	85.05[1,2,4,7,8]	69.47[1–3,5-8]	86.45[1,2,4,6-8]	83.11[1,2,4,5,7,8]	79.36[1,3-6,8]	88.71[2–7]
SD	10.75	7.90	9.29	8.55	8.09	8.13	7.57	8.07	7.86
n	2,772	522	456	363	465	283	152	142	389
Percentile									
5	613.74	626.50	625.91	629.19	589.09	627.84	659.28	584.04	618.89
10	639.55	652.85	652.53	656.32	610.68	645.63	685.51	597.96	643.81
15	657.30	668.63	670.30	675.41	627.77	662.19	695.00	610.45	658.03
20	671.89	678.19	687.19	696.35	640.59	681.79	701.61	624.94	672.86
25	683.92	684.45	700.18	708.66	653.42	691.13	714.59	639.73	685.04
30	694.79	693.15	712.85	725.78	664.83	708.13	723.86	647.57	696.66
35	704.92	702.28	724.23	735.79	676.73	716.49	732.88	658.27	706.40
40	715.35	708.08	734.34	745.27	688.66	725.77	739.18	665.43	712.89
45	725.74	718.99	742.64	756.91	696.11	734.80	748.33	672.26	724.08
50	735.65	727.12	752.10	764.31	703.35	741.59	756.99	679.92	733.35
55	744.84	735.85	761.71	773.04	715.51	751.46	767.72	688.98	744.16
60	753.76	743.14	772.88	782.45	725.56	756.88	777.53	695.54	750.25
65	764.34	754.92	782.51	790.65	737.45	763.44	786.34	708.74	758.34
70	774.77	767.46	794.96	797.61	751.02	771.37	796.95	718.20	771.33
75	784.97	776.56	803.61	806.71	759.19	779.07	808.27	737.08	780.48
80	797.88	787.91	815.70	819.93	775.27	787.97	813.56	749.74	790.05
85	811.29	802.04	829.25	845.82	789.24	807.12	834.95	764.19	804.73
90	833.89	824.53	855.53	863.57	805.72	830.98	855.48	788.64	821.23
95	864.77	856.41	888.18	890.24	838.07	863.13	891.76	823.56	856.08
Mean	735.79	732.63[2–4,6,7]	753.18[1,4,7,8]	761.28[1,4,5,7,8]	707.37[1–3,5,6,8]	740.68[3,4,7]	762.72[1,4,7,8]	686.36[1–3,5,6,8]	733.77[2–4,6,7]
SD	76.36	67.91	78.55	77.14	77.66	68.43	69.53	71.86	71.68
n	2,772	522	456	363	465	283	152	142	389

*[1]DB = defensive back; [2]DL = defensive linemen; [3]LB = linebacker; [4]OL = offensive linemen; [5]RB = running back; [6]TE = tight end; [7]QB = quarterback; [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

recorded as the distance from the starting line to the subject's heel closest to the starting line to the nearest 1.0 inch (2.54 cm).

Pro Agility. To perform the PA, subjects began in a 3-point stance facing forward with their hand on the ground. Subjects then turned 90° to the right and ran 5 yds (4.57 m), touched a line on the turf with their right hand, turned 180° to the left and ran 10 yds, touched a line on the turf with their left hand, and turned back 180° to the right and ran 5 yds through the starting line. The test began when subject's hand broke contact with the ground and ended when they ran through the starting line at the end. These times were recorded using laser timing gates (Zybek Sports).

L-Cone Drill. To perform the LC drill, subjects began in a 3-point stance facing forward with their hand on the ground. Subjects then sprinted forward to a cone 5 yds away (4.57 m), touched the cone with their right hand, turned 180° and sprinted back to the starting line, turned 180° and sprinted toward the second cone, once past the cone, turned 90° to the right and sprinted toward a cone 5 yds away, circled around the third cone in a counterclockwise movement, sprinted back around the second cone, and finished across the starting line. The test began when subject's hand broke contact with the ground and ended when they ran through the starting line at the end. These times were recorded using laser timing gates (Zybek Sports).

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Table 6

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) broad jump distance (in centimeters).*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	248.92	284.35	243.84	271.78	231.14	276.48	273.56	256.54	283.59
10	256.54	289.56	254.00	279.40	238.76	281.18	279.40	261.62	287.02
15	264.16	292.10	256.54	281.94	243.84	284.48	281.94	266.70	292.10
20	271.78	294.64	261.62	284.48	246.38	287.02	283.97	269.24	294.64
25	276.86	299.09	264.16	287.02	248.92	289.56	285.75	275.59	297.18
30	281.94	299.72	269.24	289.56	251.46	292.10	287.02	279.40	299.72
35	284.48	302.26	271.78	292.10	254.00	294.64	289.56	281.18	302.26
40	289.56	304.80	276.86	294.64	256.54	297.18	291.08	283.97	304.80
45	292.10	304.80	279.40	297.18	259.08	299.72	292.10	284.48	304.80
50	294.64	307.34	281.94	298.45	261.62	299.72	294.64	287.02	307.34
55	297.18	309.88	284.48	299.72	264.16	302.26	297.18	287.02	309.88
60	299.72	309.88	287.02	302.26	266.70	304.80	297.18	289.56	312.42
65	302.26	312.42	289.56	304.80	269.24	307.34	299.72	292.10	312.42
70	304.80	314.96	292.10	307.34	271.78	307.34	302.26	294.64	314.96
75	307.34	317.50	294.64	309.88	274.32	309.88	304.80	294.64	317.50
80	312.42	320.04	299.21	312.42	276.86	312.42	307.34	299.72	320.04
85	314.96	325.12	302.26	314.96	279.40	317.50	307.34	302.26	322.58
90	320.04	330.20	304.80	317.50	284.48	322.58	312.42	307.34	327.66
95	327.66	335.41	313.56	327.66	289.56	330.20	318.26	312.42	333.63
Mean	291.85	308.33[2–7]	280.37[1,3-6,8]	298.56[1,2,4,7,8]	260.22[1–3,5-8]	300.64[1,2,4,6-8]	295.12[1,2,4,5,7,8]	285.38[1,3-6,8]	307.67[2–7]
SD	23.87	15.77	21.04	16.29	17.36	15.95	14.44	16.39	15.61
n	2,750	518	450	360	460	276	153	141	392
Percentile									
5	1,136.09	1,149.38	1,153.83	1,166.82	1,093.36	1,161.29	1,202.03	1,092.49	1,136.76
10	1,165.34	1,176.40	1,180.12	1,205.35	1,120.71	1,180.93	1,226.63	1,107.70	1,166.80
15	1,185.24	1,192.05	1,198.60	1,222.43	1,152.52	1,198.77	1,238.63	1,123.08	1,181.97
20	1,200.36	1,205.14	1,214.16	1,233.07	1,160.83	1,209.66	1,251.75	1,142.19	1,201.33
25	1,211.61	1,214.24	1,227.56	1,248.97	1,178.81	1,218.94	1,258.55	1,154.16	1,210.84
30	1,223.92	1,223.96	1,239.66	1,257.25	1,188.26	1,227.60	1,264.46	1,169.62	1,223.82
35	1,234.43	1,230.39	1,251.36	1,267.30	1,197.67	1,236.27	1,273.18	1,177.27	1,235.54
40	1,243.79	1,238.86	1,260.60	1,273.93	1,206.98	1,246.63	1,281.15	1,190.96	1,242.95
45	1,254.03	1,248.08	1,272.36	1,281.84	1,217.99	1,256.43	1,288.29	1,199.02	1,255.05
50	1,263.42	1,253.79	1,282.59	1,292.85	1,231.04	1,266.29	1,295.71	1,208.16	1,265.20
55	1,272.35	1,261.46	1,293.22	1,301.73	1,242.91	1,272.45	1,303.82	1,216.83	1,275.57
60	1,281.27	1,270.03	1,302.78	1,308.64	1,250.98	1,279.39	1,308.60	1,226.68	1,283.66
65	1,292.19	1,277.92	1,314.49	1,317.34	1,260.82	1,285.07	1,314.28	1,231.54	1,295.67
70	1,302.99	1,288.07	1,325.83	1,327.80	1,270.10	1,296.57	1,330.13	1,244.14	1,304.66
75	1,314.14	1,300.64	1,333.36	1,340.91	1,281.64	1,304.71	1,340.33	1,256.20	1,313.71
80	1,326.73	1,312.86	1,349.65	1,353.58	1,295.17	1,320.31	1,349.74	1,274.11	1,324.76
85	1,341.31	1,328.86	1,360.76	1,361.04	1,313.40	1,335.80	1,362.36	1,296.06	1,337.39
90	1,358.75	1,352.00	1,378.43	1,378.89	1,337.47	1,351.42	1,375.40	1,308.64	1,348.06
95	1,387.47	1,380.09	1,404.58	1,416.91	1,357.73	1,372.16	1,403.39	1,323.52	1,385.75
Mean	1,262.67	1,259.07[2–4,6,7]	1,281.58[1,4,5,7,8]	1,292.28[1,4,5,7,8]	1,229.21[1–3,5-8]	1,263.87[2–4,6,7]	1,299.12[1,4,5,7,8]	1,206.68[1–6,8]	1,262.89[2–4,6,7]
SD	76.99	68.72	77.98	70.97	80.50	65.85	63.09	71.57	74.67
n	2,750	518	450	360	460	276	153	141	392

*[1]DB = defensive back; [2]DL = defensive linemen; [3]LB = linebacker; [4]OL = offensive linemen; [5]RB = running back; [6]TE = tight end; [7]QB = quarterback; [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

Statistical Analyses

Descriptive statistics (mean, *SD*, and percentile rankings) were computed for all absolute and allometrically scaled performance measures. These percentile ranks were used to generate test-specific and position-specific reference values for absolute and allometrically scaled measures. One-way analyses of variances with post hoc Bonferroni's corrections were used to test for any differences for absolute and allometrically scaled performance measures across position groups.

Relationships between the body mass vs. absolute performance measures were examined using polynomial regression analyses to examine the model of best fit to determine if allometric scaling was appropriate over ratio scaling. Using X = 10-yd split, 20-yd split, 40-yd dash, bench press test, VJ, BJ, PA, or LC, Y = body mass, and a_0, a_1, a_2, and a_3 = statistically determined regression coefficients, these models were derived as follows:

$$Y = a_0 + a_1X \text{ (linear model)}$$

$$Y = a_0 + a_1X + a_2X^2 \text{ (quadratic model)}$$

$$Y = a_0 + a_1X + a_2X^2 + a_3X^3 \text{ (cubic model).}$$

The statistical significance ($p \leq 0.05$) for the increment in the proportion of variance that would be accounted for by a higher-degree polynomial (i.e., *F*-test and R^2 change in SPSS) were

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Table 7

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) pro agility time (in seconds).*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	4.94	4.52	4.98	4.69	5.12	4.59	4.73	4.69	4.56
10	4.82	4.46	4.84	4.59	5.03	4.56	4.66	4.57	4.49
15	4.74	4.43	4.79	4.56	4.99	4.52	4.60	4.54	4.45
20	4.66	4.40	4.75	4.52	4.95	4.47	4.58	4.53	4.41
25	4.60	4.38	4.72	4.50	4.91	4.45	4.53	4.49	4.38
30	4.56	4.34	4.69	4.47	4.89	4.42	4.51	4.46	4.35
35	4.51	4.33	4.65	4.44	4.87	4.40	4.50	4.44	4.33
40	4.47	4.30	4.63	4.41	4.84	4.38	4.47	4.41	4.32
45	4.44	4.28	4.60	4.39	4.81	4.35	4.45	4.40	4.30
50	4.40	4.26	4.58	4.37	4.78	4.34	4.43	4.40	4.28
55	4.38	4.25	4.56	4.36	4.76	4.33	4.40	4.36	4.26
60	4.35	4.22	4.53	4.33	4.74	4.31	4.38	4.33	4.25
65	4.32	4.20	4.50	4.31	4.71	4.29	4.37	4.32	4.21
70	4.29	4.19	4.47	4.29	4.70	4.27	4.34	4.31	4.20
75	4.26	4.17	4.44	4.27	4.68	4.25	4.33	4.28	4.18
80	4.23	4.14	4.40	4.25	4.63	4.22	4.31	4.25	4.15
85	4.20	4.12	4.37	4.22	4.60	4.20	4.28	4.21	4.13
90	4.15	4.07	4.32	4.19	4.55	4.15	4.23	4.15	4.09
95	4.09	4.02	4.23	4.14	4.49	4.10	4.20	4.10	4.07
Mean	4.45	$4.27^{2–7}$	$4.58^{1,3-8}$	$4.39^{1,2,4,6,8}$	$4.79^{1–3,5-8}$	$4.35^{1,2,4,6,8}$	$4.45^{1–5,7,8}$	$4.38^{1,2,4,6,8}$	$4.29^{2–7}$
SD	0.26	0.15	0.22	0.17	0.20	0.15	0.19	0.17	0.16
n	2,658	503	430	339	450	267	154	136	371
Percentile									
5	1.647	1.643	1.640	1.623	1.667	1.636	1.626	1.647	1.641
10	1.623	1.619	1.608	1.599	1.648	1.616	1.602	1.624	1.623
15	1.606	1.608	1.590	1.584	1.632	1.605	1.582	1.602	1.607
20	1.595	1.598	1.578	1.571	1.622	1.597	1.570	1.589	1.599
25	1.584	1.585	1.566	1.564	1.608	1.587	1.558	1.585	1.587
30	1.574	1.577	1.555	1.557	1.600	1.578	1.552	1.581	1.577
35	1.565	1.570	1.549	1.548	1.589	1.561	1.546	1.569	1.569
40	1.557	1.563	1.543	1.537	1.583	1.552	1.540	1.563	1.561
45	1.550	1.556	1.536	1.528	1.576	1.543	1.530	1.554	1.554
50	1.543	1.549	1.528	1.521	1.565	1.535	1.521	1.549	1.548
55	1.535	1.539	1.521	1.513	1.559	1.526	1.513	1.540	1.539
60	1.526	1.529	1.515	1.505	1.553	1.521	1.505	1.533	1.532
65	1.518	1.520	1.508	1.500	1.547	1.517	1.501	1.527	1.524
70	1.511	1.514	1.499	1.492	1.539	1.511	1.496	1.519	1.516
75	1.502	1.506	1.492	1.485	1.531	1.504	1.489	1.508	1.509
80	1.494	1.496	1.482	1.475	1.522	1.497	1.480	1.500	1.501
85	1.484	1.488	1.474	1.468	1.511	1.492	1.470	1.487	1.489
90	1.471	1.476	1.455	1.452	1.496	1.479	1.458	1.472	1.477
95	1.449	1.458	1.435	1.435	1.477	1.456	1.435	1.442	1.458
Mean	1.545	$1.548^{2–6}$	$1.531^{1,4,7,8}$	$1.526^{1,4,5,7,8}$	$1.570^{1–3,5-8}$	$1.543^{1,3,4,7,8}$	$1.529^{1,4,7,8}$	$1.548^{2–6}$	$1.549^{2–6}$
SD	0.061	0.057	0.062	0.058	0.061	0.055	0.064	0.061	0.059
n	2,658	503	430	339	450	267	154	136	371

*[1]DB = defensive back; [2]DL = defensive linemen; [3]LB = linebacker; [4]OL = offensive linemen; [5]RB = running back; [6]TE = tight end; [7]QB = quarterback; [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

determined using the *F*-test described by Pedhazur (22). The models of best fit for the relationships between body mass and absolute performance measures were as follows: 10-yd split—linear ($r^2 = 0.666$, $p < 0.001$), 20-yd split—quadratic ($r^2 = 0.715$, $p < 0.001$), 40-yd dash—quadratic ($r^2 = 0.757$, $p < 0.001$), bench press test—cubic ($r^2 = 0.429$, $p < 0.001$), VJ—quadratic ($r^2 = 0.432$, $p < 0.001$), BJ—quadratic ($r^2 = 0.499$, $p < 0.001$), PA—quadratic ($r^2 = 0.549$, $p = 0.019$), and LC—linear ($r^2 = 0.622$, $p < 0.001$). Therefore, all absolute performance metrics were normalized through allometric scaling in lieu of ratio scaling.

Each dependent variable was allometrically scaled for body mass. The allometric scaling procedure involved the following equation:

$$a = \frac{T}{m^b}$$

where a = allometric-scaled performance measure, T = absolute performance measure, m = body mass, and b = allometric parameter (9,20,33). The calculated allometric parameters (b) were taken as the slopes of the linear regression lines between log-transformed body mass and log-transformed performance measures (9,20,33). Because allometric scaling normalizes performance metrics to body mass, there should be at least a decrease in the magnitude of relationship between body mass and performance, if not a statistical change, if allometric scaling is appropriate. Therefore, Pearson's product moment correlation coefficients were calculated for the relationship between body mass and absolute performance

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Table 8

Percentile ranks, mean ± *SD*, and *n* sizes for raw (top) and allometrically scaled (bottom) L-cone drill time (in seconds).*†

	All	DB	DL	LB	OL	RB	TE	QB	WR
Percentile									
5	8.06	7.32	8.05	7.59	8.35	7.55	7.59	7.59	7.38
10	7.90	7.25	7.95	7.46	8.25	7.44	7.53	7.50	7.29
15	7.75	7.19	7.90	7.39	8.15	7.37	7.44	7.47	7.23
20	7.63	7.15	7.80	7.37	8.11	7.30	7.38	7.40	7.19
25	7.53	7.12	7.73	7.31	8.07	7.26	7.35	7.32	7.14
30	7.44	7.09	7.67	7.26	8.01	7.23	7.33	7.28	7.10
35	7.36	7.07	7.62	7.24	7.98	7.21	7.31	7.26	7.07
40	7.29	7.03	7.57	7.20	7.94	7.19	7.27	7.21	7.06
45	7.24	7.02	7.53	7.18	7.90	7.17	7.24	7.18	7.03
50	7.20	7.00	7.50	7.15	7.87	7.13	7.21	7.15	7.00
55	7.15	6.96	7.45	7.12	7.84	7.12	7.19	7.11	6.97
60	7.11	6.95	7.42	7.09	7.80	7.09	7.17	7.09	6.95
65	7.07	6.92	7.37	7.07	7.75	7.07	7.15	7.07	6.92
70	7.03	6.89	7.32	7.05	7.71	7.02	7.12	7.04	6.90
75	7.00	6.86	7.26	7.01	7.65	7.00	7.10	7.00	6.87
80	6.95	6.83	7.22	6.97	7.61	6.95	7.05	6.96	6.82
85	6.90	6.81	7.15	6.93	7.56	6.90	7.00	6.95	6.77
90	6.85	6.75	7.06	6.88	7.50	6.85	6.97	6.90	6.74
95	6.76	6.69	6.98	6.81	7.39	6.81	6.91	6.82	6.65
Mean	7.29	$7.00^{2–6}$	$7.51^{1,4-8}$	$7.16^{1,2,4,8}$	$7.87^{1–3,5-8}$	$7.14^{1,2,4,6,8}$	$7.23^{1,2,4,5,8}$	$7.17^{1,2,4,8}$	$7.01^{2–6}$
SD	0.40	0.20	0.33	0.23	0.30	0.22	0.20	0.23	0.22
n	2,604	491	429	338	436	259	150	130	371
Percentile									
5	2.644	2.638	2.615	2.612	2.689	2.644	2.579	2.668	2.652
10	2.609	2.600	2.590	2.566	2.655	2.609	2.546	2.629	2.606
15	2.587	2.586	2.558	2.544	2.636	2.581	2.526	2.598	2.587
20	2.568	2.568	2.545	2.520	2.616	2.569	2.504	2.580	2.570
25	2.552	2.550	2.531	2.508	2.605	2.554	2.494	2.557	2.553
30	2.538	2.537	2.521	2.497	2.592	2.542	2.483	2.546	2.542
35	2.525	2.526	2.508	2.486	2.577	2.533	2.477	2.532	2.529
40	2.513	2.517	2.495	2.476	2.564	2.514	2.471	2.521	2.517
45	2.503	2.509	2.480	2.464	2.556	2.509	2.465	2.507	2.510
50	2.493	2.501	2.470	2.453	2.545	2.497	2.455	2.498	2.501
55	2.482	2.495	2.460	2.444	2.530	2.490	2.437	2.478	2.487
60	2.472	2.486	2.453	2.435	2.517	2.482	2.429	2.470	2.478
65	2.462	2.478	2.442	2.424	2.505	2.469	2.423	2.458	2.467
70	2.450	2.470	2.427	2.418	2.492	2.459	2.411	2.451	2.458
75	2.438	2.461	2.413	2.406	2.480	2.447	2.403	2.439	2.445
80	2.425	2.447	2.404	2.395	2.466	2.436	2.392	2.433	2.434
85	2.410	2.434	2.389	2.381	2.447	2.423	2.381	2.424	2.425
90	2.391	2.418	2.361	2.367	2.430	2.398	2.373	2.407	2.401
95	2.364	2.397	2.347	2.333	2.399	2.372	2.347	2.390	2.372
Mean	2.497	$2.507^{2–4,6}$	$2.474^{1,4,5,7,8}$	$2.460^{1,4,5,7,8}$	$2.543^{1–3,5-8}$	$2.503^{2–4,6}$	$2.452^{1,4,5,7,8}$	$2.504^{2–4,6}$	$2.503^{2–4,6}$
SD	0.086	0.074	0.087	0.079	0.089	0.078	0.068	0.083	0.084
n	2,604	491	429	338	436	259	150	130	371

*[1]DB = defensive back, [2]DL = defensive linemen, [3]LB = linebacker, [4]OL = offensive linemen, [5]RB = running back, [6]TE = tight end, [7]QB = quarterback, [8]WR = wide receiver.
†Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

measures and between body mass and allometrically scaled performance measures. The following qualitative evaluations of the strength of association were made according to Mukaka (16) based on the absolute values of correlation coefficients: 0.90–1.00 = very high, 0.70–0.89 = high, 0.50–0.69 = moderate, 0.30–0.49 = low, and 0.00–0.29 = negligible. Statistical changes in the correlation coefficients before and after allometric scaling were calculated using *z*-score transformations from publicly available software (23). All other statistical analyses were performed in IBM SPSS version 28 (Chicago, IL). An alpha level of $p \leq 0.05$ was considered statistically significant.

Results

Means, *SD*s, and percentile rankings for absolute and allometrically scaled performance measures can be found in Tables 1–8, whereas calculated allometric parameters (*b*) can be found in Table 9.

Body mass exhibited moderate to high significant relationships with absolute performance measures ($|r| \geq 0.638$, $p < 0.001$) and negligible nonsignificant relationships with all allometrically scaled performance measures ($|r| \leq 0.031$, $p \geq 0.095$) except the bench press test, which exhibited a statistically significant, albeit negligible, relationship ($r = -0.053$, $p = 0.008$). The correlation

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Table 9

Calculated test-specific allometric parameters (*b*).

Test	Allometric parameter
10-yd split	0.2810
20-yd split	0.2822
40-yd dash	0.2950
Bench press	1.2250
Vertical jump	−0.4685
Broad jump	−0.3132
Pro agility	0.2258
L-cone	0.2285

coefficients for the relationships between body mass and all performance measures significantly decreased with allometric scaling ($|z\text{-score}| \geq 25.115$, $p < 0.001$).

There were significant position-related differences for body mass and the absolute performance measures (Tables 1–8). For body mass, OL > DL > TE > LB > QB, RB > WR > DB ($p \leq 0.014$). For 10- and 20-yd splits, DB, WR < RB < LB < TE, QB < DL < OL ($p < 0.001$). For 40-yd dash, DB, WR < RB < LB < TE < QB < DL < OL ($p \leq 0.003$). For the bench press test, DL, OL > LB > RB, TE > DB, WR ($p \leq 0.001$). For VJ and BJ, DB, WR > RB, LB, TE > QB, DL > OL ($p \leq 0.013$, note: RB > TE, $p \leq 0.038$). For PA, DB, WR < RB, LB, QB < TE < DL < OL ($p \leq 0.034$). For LC, DB, WR < RB, LB, QB, TE < DL < OL ($p < 0.001$, note: RB < TE, $p = 0.017$).

There were significant position-related differences for the allometrically scaled performance measures (Tables 1–8). For 10-yd split, LB, DL, TE, RB < WR, DB < OL < QB ($p \leq 0.025$), note: RB = WR ($p = 0.137$). For 20-yd split, LB, DL, TE, RB < WR, DB < QB < OL ($p \leq 0.006$, note: RB = WR, $p = 0.371$). For 40-yd dash, TE, LB, DL, RB < WR < DB < OL < QB ($p \leq 0.004$, note: RB = WR, $p = 0.053$). For the bench press test, RB, LB > DL > DB, TE > OL, WR ($p \leq 0.008$, note: TE = OL, $p = 1.000$). For VJ, TE, LB, DL, RB > WR, DB > OL, QB ($p \leq 0.004$, note: LB > RB, $p = 0.012$, note: RB = WR, DB, $p = 1.000$). For BJ, TE, LB, DL > RB, WR, DB > OL > QB ($p \leq 0.043$). For PA, LB, TE, DL, RB < DB, QB, WR < OL ($p \leq 0.013$, note LB < RB, $p = 0.010$). For LC, TE, LB, DL < WR, RB, QB, DB < OL ($p \leq 0.009$).

Discussion

The primary results of this study provide updated percentile rank values for athletic performance measures from the NFL Scouting Combine, with the addition of the 10- and 20-yd splits, which have not been previously published, expressed in absolute and allometrically scaled terms. Based on the results of the polynomial regression analyses in this study, and the decrease in relationships between body mass and performance measures after allometric scaling, these results support the use of allometric scaling over ratio scaling because of the lack of consistent linear relationships between body mass and absolute performance measures, with the only variables best characterized with a linear relationship being the 10-yd split and LC. This study also demonstrates position-specific differences for absolute and allometrically scaled performance variables, which may provide further insight regarding the role that body mass plays in athletic performance across different positions in American football. All in all, coaches and practitioners may use the calculated percentile rankings and allometric scaling parameters to assess absolute and normalized athletic performance capabilities of NFL Scouting Combine subjects or for American football athletes desiring to compete at the highest level of competition.

Only 1 previous study has published percentile rank values for NFL Scouting Combine performance measures (20). Nuzzo (20) published percentile rankings for absolute measures of the 40-yd dash, bench press test, VJ, BJ, PA, and LC for NFL Scouting Combine subjects from 1999 to 2014. However, since Fitzgerald and Jensen (5) demonstrated that NFL Scouting Combine performance metrics significantly increased from 1999–2000 to 2015–2016, it may be pertinent to provide updated percentile rankings within a narrower window of time reflective of the current generation of players. Thus, this study contains up-to-date percentile rankings, including the 10- and 20-yd splits, which have never been published previously, using data from the 2015–2019 NFL Scouting Combines, which may more accurately reflect performance characteristics of the current generation of elite collegiate and professional American football players. Furthermore, several previous studies have recommended using allometric scaling for assessing normalized performance measures for NFL Scouting Combine subjects and athletes among various sports at different levels of competition (4,6,20,21,27,29,31). In fact, Nuzzo (20) stated, "the literature is still scant in basic descriptive statistics (e.g., percentile ranks) for allometric-scaled performance measures." To address this concern, this study includes percentile rankings for performance measures allometrically scaled for body mass and the allometric parameters by which this scaling was achieved. Thus, the present data provide yet another method by which coaches and practitioners may quantify and compare athletic performance measures for NFL Scouting Combine subjects or potential subjects.

Because one of the underlying assumptions for ratio scaling (dividing absolute performance by body mass) is a linear model of best fit between 2 variables (9,17,18), the appropriateness of allometric scaling in lieu of ratio scaling was tested for through polynomial regression analyses to determine the model of best fit for the relationship between body mass and absolute performance measures. The results demonstrated no consistency in the models of best fit, which ranged from no significant models to linear, quadratic, and cubic models. In fact, the only variables that were best characterized by a linear relationship were the 10-yd split and LC, indicating that ratio scaling may potentially be appropriate for these tests but not others. The use of allometric scaling is further supported by the decrease in correlation coefficients, both in terms of significance and magnitude of relationships, for the relationships between body mass and performance measures before and after allometric scaling. Thus, when attempting to compare normalized NFL Scouting Combine performance measures, the present results suggest that allometric scaling provides an appropriate method by which performance may be normalized. This may become important for coaches and practitioners who desire to examine position-specific differences in athletic performance because body mass changes with position, which may be valuable in determining the best strength and conditioning principles to follow for specific positions in American football.

In this study, there were position-specific differences for absolute performance measures such that skill positions (DB and WR) tended to be faster, jump higher and farther, and have greater change-of-direction capabilities than big skill positions (LB, QB, RB, TE), who performed better in these respects to linemen (OL and DL). By contrast, for the bench press test, the results were the exact opposite, with linemen > big skill > skill positions. Despite the plethora of studies examining NFL

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Scouting Combine metrics (2,3,5,7,20,24–30,32), only 2 have provided position-specific statistical comparisons (5,7). Fitzgerald and Jensen (5) and Gillen et al. (7) found similar results compared with this study, with skill position groups having the best overall absolute performance, followed by big skill position groups, then linemen. These results are in agreement with similar comparisons in high school–aged American football players (11,14,15), which may speak to the unique position-specific differences that span across players of various levels of competition. Interestingly, Gillen et al. (7) noted that body mass was one of the variables with the greatest discrepancy between NFL Scouting Combine subjects and high school scouting combine subjects, which may speak to the necessity of future studies to consider body mass when assessing differences in performance across skill levels. Nevertheless, although position-specific differences were evident for absolute performance measures, the nature of these differences changed when performance was allometrically scaled.

No previous studies have reported position-specific differences for allometrically scaled performance measures from NFL Scouting Combine subjects. Allometric scaling revealed that DL and most big skill position groups (LB, RB, and TE) tended to have superior performance than skill position groups (DB and WR), with QB and OL having the poorest allometrically scaled performance. Thus, it seems that a greater understanding of position-specific differences in athletic performance may be gained by considering differences in body mass and potentially underlying muscle mass. Using air displacement plethysmography, Provencher et al. (24) quantified body fat percentage among NFL Scouting Combine subjects, revealing that for body fat percentage, OL > DL > QB > TE > LB > RB > DB > WR, which follows the reverse pattern of absolute performance differences from this study. Based on their results, when quantifying fat-free mass (FFM, kg), the results of Provencher et al. (24) present the following: OL > DL > TE > LB > RB > QB > WR > DB. Thus, DL, LB, RB, and TE, all of which had the greatest allometrically scaled performance in this study, had greater FFM than DBs and WRs, who achieved the highest absolute performance metrics. In another study examining body composition among Division I collegiate American football players using dual-energy x-ray absorptiometry, Bosch et al. (1) reported that not only did DL, LB, RB, and TE have greater FFM than all positions except OL but also these positions had greater FFM of the lower body. Thus, it is possible that the greater allometrically scaled performance for these positions is because of the greater overall and lower-body–specific FFM, an indicator of muscle mass, which aids in executing fast powerful movements, particularly given the demands of those positions. Based on the results of Provencher et al. (24) and Bosch et al. (1), one may question why OL do not have the greatest allometrically scaled performance despite tending to have the greatest overall quantity of FFM. This may be the result of the nature of the position, which allows better success with greater overall body mass, while not requiring as great of a degree movements such as sprinting, jumping, and change of direction, all of which are assessed during the NFL Scouting Combine. Offensive linemen typically have limited movement during a given play in American football, whereas DL, LB, RB, and TE regularly sprint, change direction, and engage in tackling maneuvers more than other positions (1,25). Nevertheless, further research examining the effects of allometrically scaling athletic performance to FFM compared with overall body mass are necessary to test this hypothesis and better understand the effects that body size, and potentially FFM, has on athletic performance across American football positions.

In conclusion, this study provides new updated percentile rankings, which researchers and practitioners should consider revisiting and updating at regular time intervals because of potential changes in performance across time (5). Furthermore, this is the first study to present percentile rankings for allometrically scaled performance, providing coaches and practitioners normalized data to compare future NFL Scouting Combine subjects against. The present results also demonstrated position-specific differences for absolute performance measures, which were modified with allometric scaling such that skill positions achieved the best absolute performance while big skill positions achieved the best allometrically scaled performance. This provides unique insight regarding the role that body mass, and potentially FFM (1,24), may play in position-specific differences of athletic performance, although further research examining how body mass and FFM differentially influence allometric scaling parameters is necessary. Nevertheless, coaches and practitioners may use the percentile rankings and allometric parameters from this study to assess athletic performance among NFL Scouting Combine subjects, providing a gauge of absolute and normalized athletic performance.

Practical Applications

Coaches and practitioners who seek to determine the performance level of NFL Scouting Combine subjects should use normative reference values (percentile rankings) that allow comparisons to recent NFL Scouting Combine subjects. Furthermore, many coaches and practitioners seek not only to examine absolute performance measures but also to normalize performance to body mass to allow comparisons of players of different positions and performance levels. For example, if a WR jumps higher than a DL, yet the DL has greater normalized athletic performance, it becomes evident that the only reason differences in jump height exist because of the confounding effect of body mass. This suggests that although the WR jumped higher, the DL likely has greater power output. It is possible that this discrepancy in normalized jump height, and potentially power output, may be the result of differences in underlying muscle mass, providing greater information regarding position-related differences not found by assessments of absolute performance. These types of comparisons of normalized performance measures may help guide strength and conditioning professionals regarding position-specific strength and conditioning programs for American football players. This study provides the necessary equation and allometric parameters by which coaches and practitioners may normalize NFL Scouting Combine data effectively for future evaluations of normalized athletic performance among elite American football players.

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